

LangChain & Agentic AI – From Basics to Advanced

demo- 5 Notes

Understanding Messages and Message Types:

Message: It is a structured unit of communication in a conversation with a chat model.

3 types of Messages:

1. SystemMessage--->Instructions/context that guide the model. It decides behaviour of the Model.
2. HumanMessage--->Message from human/user
3. AIMessage--->Response from Model

Message Object vs content:

print(message) -->It prints message object representation |
Complete Message object

`print(message.content)` -->It prints specific content of the message | Actual Message content

```
from langchain_core.messages import SystemMessage,  
HumanMessage, AIMessage
```

```
system_message = SystemMessage(content = "You are  
experienced Python Teacher")
```

```
human_message = HumanMessage(content = "What is  
Python?")
```

```
ai_message = AIMessage(content = "Python is a wonderful  
programming language")
```

```
#print(system_message)
```

```
#print(system_message.content)
```

```
#print(human_message)
```

```
#print(human_message.content)
```

```
print(ai_message)
```

```
print(ai_message.content)
```

```
[  
    system message,  
    human message,  
    ai message,  
    human message,  
    ai message,  
    human message  
    ....  
]
```

```
import os # os is module
```

```
from langchain_openai import ChatOpenAI
```

```
from langchain_core.messages import  
SystemMessage, HumanMessage
```

```
OPENAI_API_KEY = os.getenv("OPENAI_API_KEY") #function
```

```
llm = ChatOpenAI(model="gpt-4o",api_key=OPENAI_API_KEY)
```

```
system_message = SystemMessage(content="You are  
Experienced Python Trainer")
```

```
human_message = HumanMessage(content="Explain about  
Python in a single line")
```

```
ai_msg = llm.invoke([system_message,human_message])
```

```
#print(type(ai_msg))
```

```
print(ai_msg)
```

```
#print(ai_msg.content)
```

```
system_message = SystemMessage(content="You are  
Experienced Life Skills Coach")
```

```
human_message = HumanMessage(content="Explain about  
Law of Attraction")
```

```
#ai_msg = llm.invoke([system_message,human_message])
```

```
#print(type(ai_msg))
```

```
#print(ai_msg)
```

```
#print(ai_msg.content)
```